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Information in the passage suggests that which of the following is true of carotenoids that a male animal uses for detoxification processes?

A. They were not acquired directly from plants and algae.

B. They cannot be replenished through foraging.

C. They cannot be used simultaneously to brighten coloration.

D. They do not affect the animal's susceptibility to parasites.

E. They increase the chances that the animal will be selected as a mate.

Line Linda Kerber argued in the mid-1980s that after the American Revolution (1775–1783), an ideology of "republican motherhood" resulted in a surge of educational opportunities for women in the United (5) States. Kerber maintained that the leaders of the new nation wanted women to be educated in order to raise politically virtuous sons. A virtuous citizenry was considered essential to the success of the country's republican form of government; (10) virtue was to be instilled not only by churches and schools, but by families, where the mother's role was crucial. Thus, according to Kerber, motherhood became pivotal to the fate of the republic, providing justification for an unprecedented attention to female (15) education.

Introduction of the "republican motherhood" thesis dramatically changed historiography. Prior to Kerber's work, educational historians barely mentioned women and girls; Thomas Woody's (20) 1929 work is the notable exception. Examining newspaper advertisements for academies, Woody found that educational opportunities increased for both girls and boys around 1750. Pointing to "An Essay on Woman" (1753) as reflecting a shift in (25) view, Woody also claimed that practical education for females had many advocates before the Revolution. Woody's evidence challenges the notion that the Revolution changed attitudes regarding female education, although it may have accelerated (30) earlier trends. Historians' reliance on Kerber's "republican motherhood" thesis may have obscured the presence of these trends, making it difficult to determine to what extent the Revolution really changed women's lives.

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Questions 527–531 refer to the passage.

527. According to the passage, Kerber maintained that which of the following led to an increase in educational opportunities for women in the United States after the American Revolution?

A. An unprecedented demand by women for greater educational opportunities in the decades following the Revolution

B. A new political ideology calling for equality of opportunity between women and men in all aspects of life

C. A belief that the American educational system could be reformed only if women participated more fully in that system

D. A belief that women needed to be educated if they were to contribute to the success of the nation's new form of government

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E. A recognition that women needed to be educated if they were to take an active role in the nation's schools and churches

528. According to the passage, within the field of educational history, Thomas Woody's 1929 work was

A. innovative because it relied on newspaper advertisements as evidence

B. exceptional in that it concentrated on the period before the American Revolution

C. unusual in that it focused on educational attitudes rather than on educational practices

D. controversial in its claims regarding educational opportunities for boys

E. atypical in that it examined the education of girls

529. The passage suggests that Woody would have agreed with which of the following claims regarding "An Essay on Woman"?

A. It expressed attitudes concerning women's education that were reflected in new educational opportunities for women after 1750.

B. It persuaded educators to offer greater educational opportunities to women in the 1750s.

C. It articulated ideas about women's education that would not be realized until after the American Revolution.

D. It offered one of the most original arguments in favor of women's education in the United States in the eighteenth century.

E. It presented views about women's education that were still controversial in Woody's own time.

530. The passage suggests that, with regard to the history of women's education in the United States, Kerber's work differs from Woody's primarily concerning which of the following?

A. The extent to which women were interested in pursuing educational opportunities in the eighteenth century

B. The extent of the support for educational opportunities for girls prior to the American Revolution

C. The extent of public resistance to educational opportunities for women after the American Revolution

D. Whether attitudes toward women's educational opportunities changed during the eighteenth century

E. Whether women needed to be educated in order to contribute to the success of a republican form of government

531. According to the passage, Kerber argued that political leaders thought that the form of government adopted by the United States after the American Revolution depended on which of the following for its success?

A. Women assuming the sole responsibility for instilling political virtue in children

B. Girls becoming the primary focus of a reformed educational system that emphasized political virtue

C. The family serving as one of the primary means by which children were imbued with political virtue

D. The family assuming many of the functions previously performed by schools and churches

E. Men and women assuming equal responsibility for the management of schools, churches, and the family

Line Years before the advent of plate tectonics—the widely accepted theory, developed in the mid-1960s, that holds the major features of Earth's surface are created by the horizontal motions (5) of Earth's outer shell, or lithosphere—a similar theory was rejected by the geological community. In 1912, Alfred Wegener proposed, in a widely debated theory that came to be called continental drift, that Earth's continents were mobile. To most (10) geologists today, Wegener's *The Origin of Continents and Oceans* appears an impressive and prescient document, containing several of the essential presumptions underlying plate tectonics theory: the horizontal mobility of pieces of Earth's crust; the (15) essential difference between oceanic and continental

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(15) essential difference between oceanic and continental crust; and a causal connection between horizontal displacements and the formation of mountain chains. Yet despite the considerable overlap between Wegener's concepts and the later widely embraced (20) plate tectonics theory, and despite the fact that continental drift theory presented a possible solution to the problem of the origin of mountains at a time when existing explanations were seriously in doubt, in its day Wegener's theory was rejected by the vast (25) majority of geologists.

Most geologists and many historians today believe Wegener's theory was rejected because of its lack of an adequate mechanical basis. A group of researchers argues continental drift theory was (30) rejected because it did not explain how continents could move through an apparently solid oceanic floor. However, as Anthony Hallam has pointed out, many scientific phenomena, such as the ice ages, have been accepted before they could be fully (35) explained. The most likely cause for the rejection of continental drift—a cause that has been largely ignored because we consider Wegener's theory to have been validated by the theory of plate tectonics—is the nature of the evidence that was put forward (40) to support it. Most of Wegener's evidence consisted of homologies—similarities of patterns and forms based on direct observations of rocks in the field, supported by the use of hammers, hand lenses, and field notebooks. In contrast, the data supporting (45) plate tectonics were impressively geophysical: instrumental determinations of the physical properties of Earth garnered through the use of seismographs and magnetometers.

Questions 532–536 refer to the passage.

532. It can be inferred from the passage that geologists today would be most likely to agree with which of the following statements about Wegener's *The Origin of Continents and Oceans*?

A. It was a worthy scientific effort that was ahead of its time.

B. It was based on evidence that was later disproved.

C. It was directly responsible for the acceptance of the theory of plate tectonics.

D. It has been disproved by continental drift theory.

E. It misrepresented how horizontal displacements cause the formation of mountain chains.

533. The author of the passage suggests the most likely explanation for the geological community's response to continental drift theory in its day was that the theory

A. was in conflict with certain aspects of plate tectonics theory

B. failed to account for how mountains were formed

C. did not adequately explain how continents moved through the ocean floor

D. was contradicted by the geophysical data of the time

E. was based on a kind of evidence that was considered insufficiently convincing

534. The author of the passage probably refers to the "considerable overlap" (see line 18) between the continental drift theory and plate tectonics theory in order to

A. suggest that plate tectonics theory is derived from Wegener's work

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B. introduce a discussion comparing the elements of the two theories

C. examine the question of whether continental drift theory was innovative in its time

D. provide a reason why it might seem surprising that continental drift theory was not more widely embraced by geologists

E. cite an explanation that has been frequently offered for Wegener's high standing among geologists today

535. The author cites Hallam (see line 32) on the ice ages primarily in order to

A. provide an example of a geologic phenomenon with precise causes that are not fully understood by geologists today

B. criticize the geological community for an apparent lack of consistency in its responses to new theories

C. offer evidence held to undermine a common view of why Wegener's theory was not accepted in its day

D. give an example of a modern scientist who believes Wegener's theory was rejected because it failed to adequately explain the mechanical basis of continental drift

E. support some researchers' rationale for why Wegener's theory was rejected by most geologists in the early twentieth century

536. The author of the passage most likely discusses the "essential presumptions" (see lines 12–13) of *The Origin of Continents and Oceans* in order to

A. indicate features of Wegener's theory that caused it to be doubted in its day

B. show why Wegener's theory is now regarded as prescient

C. indicate differences between plate tectonics and the theory of continental drift

D. cite features of the theory of continental drift for which no evidence was available in Wegener's day

E. point out aspects of Wegener's theory that were accepted well before the advent of plate tectonics

Line Manufacturers have to do more than build large manufacturing plants to realize economies of scale. It is true that as the capacity of a manufacturing operation rises, costs per unit of output fall as plant (5) size approaches "minimum efficient scale," where the cost per unit of output reaches a minimum, determined roughly by the state of existing technology and size of the potential market. However, minimum efficient scale cannot be fully realized unless a steady (10) "throughput" (the flow of materials through a plant) is attained. The throughput needed to maintain the optimal scale of production requires careful coordination not only of the flow of goods through the production process, but also of the flow of input from (15) suppliers and the flow of output to wholesalers and final consumers. If throughput falls below a critical point, unit costs rise sharply and profits disappear. A manufacturer's fixed costs and "sunk costs" (original capital investment in the physical plant) do not (20) decrease when production declines due to inadequate supplies of raw materials, problems on the factory floor, or inefficient sales networks. Consequently, potential economies of scale are based on the physical and engineering characteristics of the (25) production facilities—that is, on tangible capital—but realized economies of scale are operational and organizational, and depend on knowledge, skills, experience, and teamwork—that is, on organized human capabilities, or intangible capital.

(30) The importance of investing in intangible capital

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(30) The importance of investing in intangible capital becomes obvious when one looks at what happens in new capital-intensive manufacturing industries. Such industries are quickly dominated, not by the first firms to acquire technologically sophisticated plants of (35) theoretically optimal size, but rather by the first to exploit the full potential of such plants. Once some firms achieve this, a market becomes extremely hard to enter. Challengers must construct comparable plants and do so after the first movers have already (40) worked out problems with suppliers or with new production processes. Challengers must create distribution networks and marketing systems in markets where first movers have all the contacts and know-how. And challengers must recruit management (45) teams to compete with those that have already mastered these functional and strategic activities.

Questions 537–541 refer to the passage.

537. The passage suggests that in order for a manufacturer in a capital-intensive industry to have a decisive advantage over competitors making similar products, the manufacturer must

A. be the first in the industry to build production facilities of theoretically optimal size

B. make every effort to keep fixed and sunk costs as low as possible

C. be one of the first to operate its manufacturing plants at minimum efficient scale

D. produce goods of higher quality than those produced by direct competitors

E. stockpile raw materials at production sites in order to ensure a steady flow of such materials

538. The passage suggests that which of the following is true of a manufacturer's fixed and sunk costs?

A. The extent to which they are determined by market conditions for the goods being manufactured is frequently underestimated.

B. If they are kept as low as possible, the manufacturer is very likely to realize significant profits.

C. They are the primary factor that determines whether a manufacturer will realize economies of scale.

D. They should be on a par with the fixed and sunk costs of the manufacturer's competitors.

E. They are not affected by fluctuations in a manufacturing plant's throughput.

539. In the context of the passage as a whole, the second paragraph serves primarily to

A. provide an example to support the argument presented in the first paragraph

B. evaluate various strategies discussed in the first paragraph

C. introduce evidence that undermines the argument presented in the first paragraph

D. anticipate possible objections to the argument presented in the first paragraph

E. demonstrate the potential dangers of a commonly used strategy

540. The passage LEAST supports the inference that a manufacturer's throughput could be adversely affected by

A. a mistake in judgment regarding the selection of a wholesaler

B. a breakdown in the factory's machinery

C. a labor dispute on the factory floor

D. an increase in the cost per unit of output

E. a drop in the efficiency of the sales network

541. The primary purpose of the passage is to

A. point out the importance of intangible capital for realizing economies of scale in manufacturing

B. show that manufacturers frequently gain a competitive advantage from investment in large manufacturing

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B. show that manufacturers frequently gain a competitive advantage from investment in large manufacturing facilities

C. argue that large manufacturing facilities often fail because of inadequate investment in both tangible and intangible capital

D. suggest that most new industries are likely to be dominated by firms that build large manufacturing plants early

E. explain why large manufacturing plants usually do not help manufacturers achieve economies of scale

Line Resin is a plant secretion that hardens when exposed to air; fossilized resin is called amber. Although Pliny in the first century recognized that amber was produced from "marrow discharged by (5) trees," amber has been widely misunderstood to be a semiprecious gem and has even been described in mineralogy textbooks. Confusion also persists surrounding the term "resin," which was defined before rigorous chemical analyses were available. (10) Resin is often confused with gum, a substance produced in plants in response to bacterial infections, and with sap, an aqueous solution transported through certain plant tissues. Resin differs from both gum and sap in that scientists have not determined a (15) physiological function for resin.

In the 1950s, entomologists posited that resin may function to repel or attract insects. Fraenkel conjectured that plants initially produced resin in nonspecific chemical responses to insect attack (20) and that, over time, plants evolved that produced resin with specific repellent effects. But some insect species, he noted, might overcome the repellent effects, actually becoming attracted to the resin. This might induce the insects to feed on those (25) plants or aid them in securing a breeding site. Later researchers suggested that resin mediates the complex interdependence, or "coevolution," of plants and insects over time. Such ideas led to the development of the specialized discipline of chemical (30) ecology, which is concerned with the role of plant chemicals in interactions with other organisms and with the evolution and ecology of plant antiherbivore chemistry (plants' chemical defenses against attack by herbivores such as insects).

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Questions 542–545 refer to the passage.

542. According to the passage, which of the following is true of plant antiherbivore chemistry?

A. Changes in a plant's antiherbivore chemistry may affect insect feeding behavior.

B. A plant's repellent effects often involve interactions between gum and resin.

C. A plant's antiherbivore responses assist in combating bacterial infections.

D. Plant antiherbivore chemistry plays only a minor role in the coevolution of plants and insects.

E. Researchers first studied repellent effects in plants beginning in the 1950s.

543. Of the following topics, which would be most likely to be studied within the discipline of chemical ecology as it is described in the passage?

A. Seeds that become attached to certain insects, which in turn carry away the seeds and aid in the reproductive cycle of the plant species in question

B. An insect species that feeds on weeds detrimental to crop health and yield, and how these insects might aid

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B. An insect species that feeds on weeds detrimental to crop health and yield, and how these insects might aid in agricultural production

C. The effects of deforestation on the life cycles of subtropical carnivorous plants and the insect species on which the plants feed

D. The growth patterns of a particular species of plant that has proved remarkably resistant to herbicides

E. Insects that develop a tolerance for feeding on a plant that had previously been toxic to them, and the resultant changes within that plant species

544. The author refers to "bacterial infections" (see line 11) most likely in order to

A. describe the physiological function that gum performs in plants

B. demonstrate that sap is not the only substance that is transported through a plant's tissues

C. explain how modern chemical analysis has been used to clarify the function of resin

D. show that gum cannot serve as an effective defense against herbivores

E. give an example of how confusion has arisen with regard to the nature of resin

545. The author of the passage refers to Pliny most probably in order to

A. give an example of how the nature of amber has been misunderstood in the past

B. show that confusion about amber has long been more pervasive than confusion about resin

C. make note of the first known reference to amber as a semiprecious gem

D. point out an exception to a generalization about the history of people's understanding of amber

E. demonstrate that Pliny believed amber to be a mineral

Line During the 1980s, many economic historians studying Latin America focused on the impact of the Great Depression of the 1930s. Most of these historians argued that although the Depression (5) began earlier in Latin America than in the United States, it was less severe in Latin America and did not significantly impede industrial growth there. The historians' argument was grounded in national government records concerning tax revenues and (10) exports and in government-sponsored industrial censuses, from which historians have drawn conclusions about total manufacturing output and profit levels across Latin America. However, economic statistics published by Latin American (15) governments in the early twentieth century are neither reliable nor consistent; this is especially true of manufacturing data, which were gathered from factory owners for taxation purposes and which therefore may well be distorted. Moreover, (20) one cannot assume a direct correlation between the output level and the profit level of a given industry, as these variables often move in opposite directions. Finally, national and regional economies are composed of individual firms and industries, (25) and relying on general, sweeping economic indicators may mask substantial variations among these different enterprises. For example, recent analyses of previously unexamined data on textile manufacturing in Brazil and Mexico suggest that the (30) Great Depression had a more severe impact on this Latin American industry than scholars had recognized.

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Questions 546–548 refer to the passage.

546. The primary purpose of the passage is to

A. compare the impact of the Great Depression on Latin America with its impact on the United States

B. criticize a school of economic historians for failing to analyze the Great Depression in Latin America within a global context

C. illustrate the risks inherent in comparing different types of economic enterprises to explain economic phenomena

D. call into question certain scholars' views concerning the severity of the Great Depression in Latin America

E. demonstrate that the Great Depression had a more severe impact on industry in Latin America than in certain other regions

547. Which of the following conclusions about the Great Depression is best supported by the passage?

A. It did not impede Latin American industrial growth as much as historians had previously thought.

B. It had a more severe impact on the Brazilian and the Mexican textile industries than it had on Latin America as a region.

C. It affected the Latin American textile industry more severely than it did any other industry in Latin America.

D. The overall impact on Latin American industrial growth should be reevaluated by economic historians.

E. Its impact on Latin America should not be compared with its impact on the United States.

548.

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